Stage 1 structure

Two batteries on one bus, with an error in each owner's terminal value

Companion to `docs/model_structure.pdf`

Figures and numbers: `scripts/make_stage1_figures.py`

28 October 2026

What stage 1 adds. Stage 0 had one battery and one error, the price forecast $\hat{\lambda}$. Stage 1 keeps the same system but splits the battery into **two batteries, A and B**, owned separately, and adds a **second error**: each owner's estimate $\hat{w}_e$ of its own end-of-horizon value w_e. Baldick's design is again the benchmark; the question is how much welfare the current design loses when two owners build their bids independently, each from its own price forecast and its own estimate of what stored energy is worth at the end of the day.

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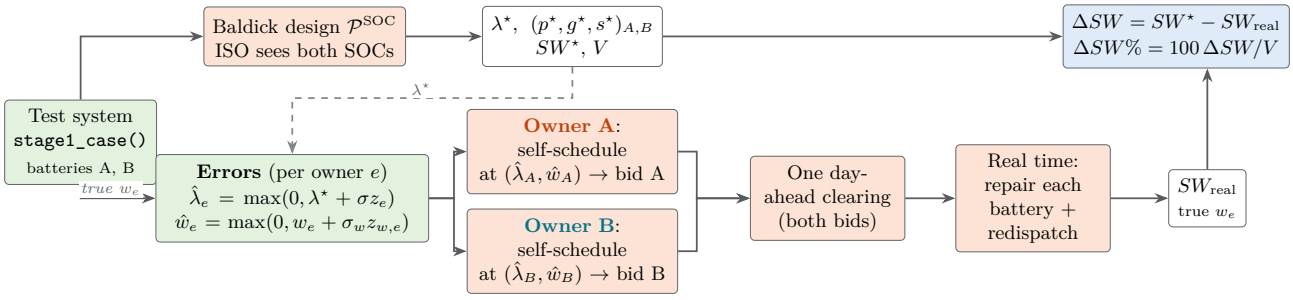
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1 Big picture

The comparison is the one of stage 0: a benchmark welfare $SW^\star$ from Baldick's design, solved once, against the realised welfare SW_{real} of the current design. What changes is the lower path: two owners now build their bids separately, each with two errors, and the two bids meet in one market.



Three facts carry over from stage 0, with one addition each:

1. **One welfare yardstick** for benchmark and realised outcome (generation cost, redispatch premium ψ, shedding at VOLL, degradation, minus $\sum_e w_e s_{e,T}$), always at the *true* w_e. A wrong $\hat{w}_e$ changes what the owner does, not how its stored energy is valued.
2. **The errors enter only the owners' own bids.** $\hat{\lambda}_e$ and $\hat{w}_e$ are used by owner e to build its bid (`B_fcst`; $\hat{w}_e$ also in `B_priv`). The system, the clearing and the welfare use true values.
3. **The benchmark is solved once.** Each draw solves two self-schedules, one clearing with both bids, and one real-time repair per battery.

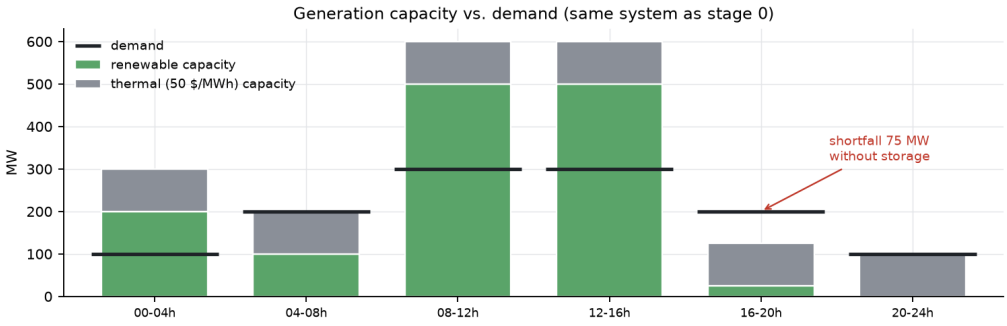
2 What is the same as stage 0

Everything below is unchanged and documented in `docs/model_structure.pdf`; it is only recalled here.

Step	What it does	`model_structure.pdf`
M1 Baldick $\mathcal{P}^{\mathrm{SOC}}$	ISO optimises dispatch and both SOCs; gives $\lambda^\star$, $\pi_e^\star$, $SW^\star$	§4, M1
M3 self-schedule	owner e's profit-maximising plan at a price vector, throughput tie-breaker $\kappa = 10^{-3}$	M3
M4 polyhedron Π	valid multipliers for that plan; `prelimit` rule picks one	M4
Quantity rule	two blocks per side, split at the planned quantity, $\pm\delta$	§4
M5 clearing	one merit-order auction per interval, now with both batteries' blocks	M5
M6/M7 real time	each battery does what its SOC allows; generators fill the gap at premium ψ	M6, M7
Price error	AR(1) shape z, size σ, autocorrelation ρ, floor at 0	§5
Seeds	common random numbers: draw d has the same shapes in every setting	§5.2

3 The new case

System. Exactly the stage-0 system (Baldick 2026, §4.7): one bus, six 4-hour intervals, a renewable block at 0 \$/MWh, one thermal block of 100 MW at 50 \$/MWh, inelastic demand, no ramp limits. Without storage the 16–20h interval is 75 MW short and sheds load at VOLL (1 000 \$/MWh).



Batteries. The 800 MWh battery of stage 0 is split into two batteries of 400 MWh each. They are identical except for their terminal value w:

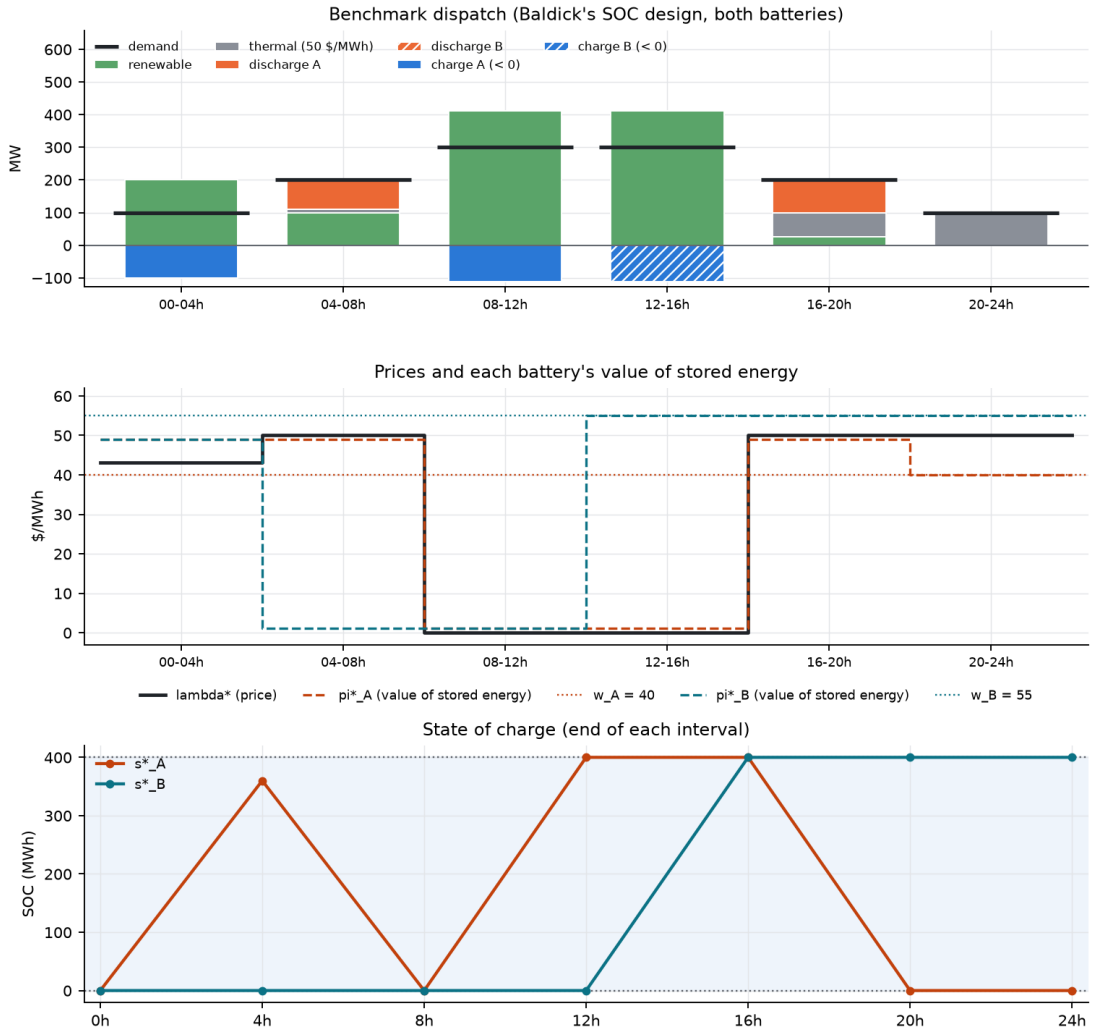
	$\bar{s}$ (MWh)	s_0	α	β	γ	$\varepsilon = \zeta$ (\$/MWh)	$\bar{p} = \bar{g}$ (MW)	w (\$/MWh)
A	400	0	0.9	1	1	1	1 000	**40**
B	400	0	0.9	1	1	1	1 000	**55**

Why this split:

- **Comparable with stage 0.** Total storage stays 800 MWh and every other parameter is Baldick's, so any difference from stage 0 comes from having two owners, not from a different system.
- **The two batteries differ only in w,** so the terminal value, and therefore its error, has a clear role. In this system the decisive number is **49 \$/MWh**: selling a stored MWh at the price of 50 earns $50 - \zeta = 49$, and the morning round trip buys at 43.1 and needs $(43.1 + \varepsilon)/\alpha = 49$ to break even. A battery with $w < 49$ sells its energy before the end of the day; with $w > 49$ it keeps it. $w_A = 40$ lies 9 below this threshold, $w_B = 55$ lies 6 above it.

Code: `cases.stage1_case(smax_each=400, w=(40, 55))`; both numbers are arguments, so other splits are one call away.

4 Baldick's result on the new case



Interval	00–04h	04–08h	08–12h	12–16h	16–20h	20–24h
Demand (MW)	100	200	300	300	200	100
Renewable used / available (MW)	200 / 200	100 / 100	411 / 500	411 / 500	25 / 25	0 / 0
Thermal at 50 (MW)	0	10	0	0	75	100
A charge / discharge (MW)	100 / –	– / 90	111 / –	–	– / 100	–
A SOC at end (MWh)	360	0	400	400	0	0
B charge / discharge (MW)	–	–	–	111 / –	–	–
B SOC at end (MWh)	0	0	0	400	400	400
$\lambda^\star$ ($/MWh)	43.1	50	0	0	50	50
$\pi_A^\star$ ($/MWh)	49	49	1.11	1.11	49	40
$\pi_B^\star$ ($/MWh)	49	1.11	1.11	55	55	55

	Stage 1 (A + B)	Stage 0 (one battery)
$SW^\star$ ($)	−17 048.89	−19 448.89
SW_{noESS} ($)	−360 000.00	−360 000.00
$V = SW^\star - SW_{\text{noESS}}$ ($)	342 951.11	340 551.11

How to read it:

- **A is the trader.** It runs the morning round trip (buy 100 MW at 43.1, sell 90 MW at 50, exactly break-even), fills up on free renewable energy at 08–12h and empties at 16–20h, where its 100 MW cover the 75 MW shortfall and replace 25 MW of thermal. Its value of stored energy $\pi_A^\star$ is 49 while it holds energy for a 50 sale, about 1 while energy is free, and $w_A = 40$ in the last interval.
- **B is the store.** It fills on free energy at 12–16h and keeps all 400 MWh to the end of the day, because

each MWh is worth $w_B = 55$ to it, more than the 49 a sale would earn. $\pi^\star_B = 55$ from the moment it is full.

- **The prices are the stage-0 prices.** $\lambda^\star = (43.1, 50, 0, 0, 50, 50)$ is unchanged.
- $SW^\star$ **is 2 400 \$ higher than in stage 0,** which is exactly 400 MWh $\times$ (55 − 49). With one battery at $w = 40$, those 400 MWh were sold at 16–24h for a net 49; here B keeps them at 55. V rises by the same amount.
- **Which battery does what is not unique.** Both are free to buy at 08–12h or 12–16h, and A could charge B's interval and vice versa: Baldick's LP returns one of several equally good splits. The welfare and the prices do not depend on that choice.

The equivalence still holds with two batteries: bids built from Baldick's own duals and quantities (`B_star`, both batteries) give $\Delta SW = 0$ to floating-point precision.

5 The two errors

5.1 Price error: the stage-0 model, now with a shared part

Each owner gets its own forecast,

$$\hat{\lambda}_{e,t} = \max\big(0,\ \lambda^\star_t + \sigma z_{e,t}\big), \qquad z_{e,t} = \sqrt{\omega}\,\chi_t + \sqrt{1-\omega}\,\eta_{e,t},$$

with χ common to both owners and η_e owner-specific (AR(1), unit variance). With one battery ω had no effect; with two it sets how alike the owners' forecasts are: $\omega = 0$ independent forecasts, $\omega = 1$ the same forecast (both buy the same data). The errors of both owners are drawn together, once per draw, so χ is truly shared.

5.2 Terminal-value error (new)

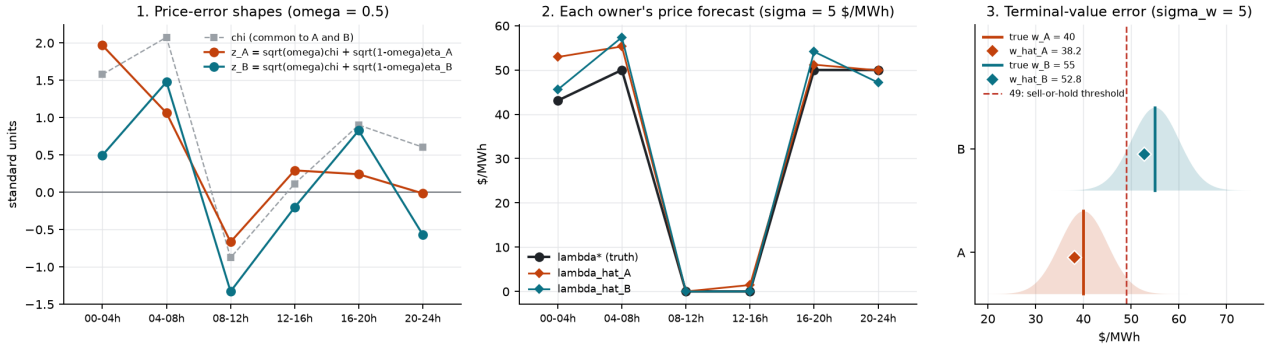
Each owner estimates what its stored energy will be worth at the end of the day:

$$\boxed{\hat{w}_e = \max\big(0,\ w_e + \sigma_w\, z_{w,e}\big), \qquad z_{w,e} \sim N(0,1) \text{ iid}}$$

one draw per battery and scenario, constant over the day, independent of the price error.

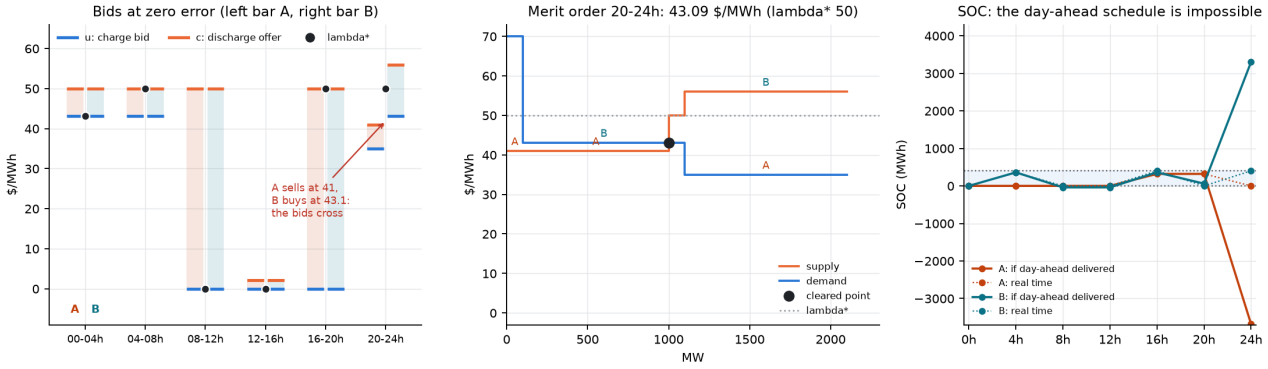
- **Why an error on** w**.** w is the owner's view of tomorrow: tomorrow's prices, its own commitments, its degradation outlook. It is as much a forecast as $\hat{\lambda}$, and Baldick's design asks the owner to submit it. In the current design it never reaches the ISO, but it shapes the bid: it is the end-point of the owner's self-schedule (M3) and fixes π_T.
- **Where it enters.** Only in the owner-built bids: `B_fcst` uses $(\hat{\lambda}_e, \hat{w}_e)$, `B_priv` uses $\hat{w}_e$. The benchmark and the welfare keep the true w_e, exactly as the price error leaves $\lambda^\star$ untouched.
- **Why it matters here.** Because of the 49 \$/MWh threshold (§3), a $\hat{w}_e$ on the wrong side of 49 flips the owner's end-of-day decision: an A with $\hat{w}_A > 49$ keeps its energy instead of covering the 16–20h shortfall; a B with $\hat{w}_B < 49$ sells energy it should have kept. At $\sigma_w = 5$ the chance of crossing is $P(z > 1.8) = 3.6\%$ for A and $P(z < -1.2) = 11.5\%$ for B.
- **Floor.** $\hat{w}_e \geq 0$, the same floor as the price: stored energy can always be kept, so it is never worth less than nothing.
- **Seeds.** $z_{w,e}$ has its own stream, `RunSettings.seed_w(d)`, so draw d has the same z_w whatever σ, ω or σ_w are (common random numbers for both errors).

One draw ($d = 0$, $\sigma = 5$, $\omega = 0.5$, $\sigma_w = 5$): the error shapes, the two forecasts, and the two terminal-value estimates ($\hat{w}_A = 38.2$, $\hat{w}_B = 52.8$, both on the correct side of 49).



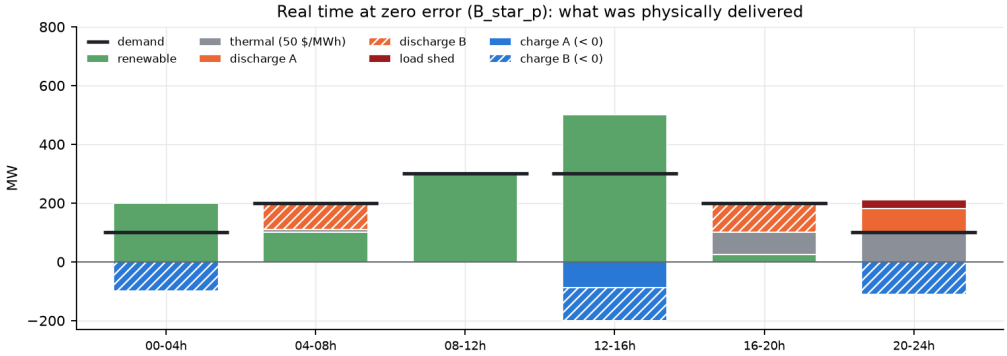
6 First check at zero error: the two bids cross

Before any error, each owner builds its bid from its own self-schedule at the *true* prices (`B_star_p`). With one battery this cost 1.30% of V. With two it costs **38.1%**, sheds 124 MWh at 20–24h, and leaves 6 976 MWh of the day-ahead schedule undeliverable.



Mechanism. The quantity rule sends two blocks per side: the planned quantity at $u+\delta$ (charge) or $c-\delta$ (discharge), and *all remaining power*, $\bar{p}-p^{\circ}$ or $\bar{g}-g^{\circ}$, at $u-\delta$ or $c+\delta$. With Baldick's non-binding power limits of 1 000 MW, this second block is up to 1 000 MW regardless of how much energy the battery holds. At 20–24h A offers to sell at $c_A = 41$ ($\pi_A = w_A = 40$ plus ζ) and B bids to buy at $u_B = 43.1$. The bids cross, so the market clears A selling 1 000 MW and B buying 900 MW at 43.09 $/MWh: the two batteries trade with each other. Neither can deliver (A is almost empty, B almost full). With one battery this cannot happen, because a battery cannot trade with itself.

In real time each battery does what its SOC allows: A delivers 80 MW, B still charges 111 MW, and the system sheds 31 MW to serve that charge (below). The real-time rule treats a battery's feasible charge as firm load, even when that means shedding; with one battery that never occurred.



This dominates every error result. A first exploratory run (40 draws, $\omega = 0.5$, current rule) gives mean losses of 21–102% of V with load shed in 52–100% of draws, and no clear pattern in σ or σ_w: the crossing, not the errors, decides the outcome. The same zero-error check under alternatives (30 draws for the error column):

Quantity rule / case	`B_star_p` loss	$\sigma = 0$, $\sigma_w = 5$: mean loss	draws shedding
Current: 2nd block = all remaining power	38.1%	22.4%	57%
Plan only (no 2nd block)	84.6%	84.6%	100%
2nd block capped by the battery's own SOC plan	**2.0%**	**2.0%**	**0%**
Current rule, $\bar{p} = \bar{g} = 100\,\text{MW}$	0.6%	0.7%	0%

"Capped by the SOC plan": the second block offers only what the battery could still absorb or deliver from the SOC its own plan reaches, $\min\big(\bar{p}, (\bar{s} - \gamma s^\circ_{t-1})/(\alpha \Delta T)\big) - p^\circ_t$ on the charge side and $\min\big(\bar{g}, (\gamma s^\circ_{t-1} - \underline{s})/(\beta \Delta T)\big) - g^\circ_t$ on the discharge side. Removing the second block altogether is worse, because those blocks are also what lets the market move energy from one battery to the other when their plans overlap (both plan to buy the same free energy).

Decisions needed before the notebook.

1. **Quantity rule.** Keep the current rule (and report the crossing as a finding of the current design with independent owners), cap the second block by the SOC plan (recommended: a battery should not offer energy it cannot have), or change the case to realistic power limits. This is the bid-curve topic that was already next in the plan.
2. **Real-time priority.** Should real time curtail a battery's charge before it sheds load? With one battery this never mattered; with two it does.

7 Proposed settings and the notebook

Settings (one CSV per setting in `results/stage1/`, 150 draws each, base seed 42, common random numbers):

Parameter	Values	Why
σ (\$/MWh)	0, 1, 2, 5, 10, 20	same as stage 0, so the curves can be laid on top of each other
σ_w (\$/MWh)	0, 2, 5, 10	from "never crosses 49" to "often crosses" for both batteries
ω	0, 0.5, 1	independent, partly shared, identical forecasts
ρ, bias, ψ, rule	0, 0, 10, `prelimit`	as in stage 0
Deterministic	`B_star`, `B_star_p`, `B_priv` ($\psi = 0, 10, 50$)	gate and references

$6 \times 4 \times 3 = 72$ stochastic settings, 10 800 draws, about 20 minutes.

Notebook `notebooks/stage_1.ipynb`, same format as stage 0: (1) the system and the two batteries, (2) Baldick's benchmark, figure and table, (3) the two errors on one draw, (4) the bids of A and B, (5) day-ahead clearing with both bids, (6) real time, (7) one draw end to end with an explorer, (8) deterministic gate (`B_star` $\Delta SW = 0$ with two batteries; $\sigma = \sigma_w = 0$ reproduces `B_star_p`), (9) results: loss over (σ, σ_w) as a heatmap, the effect of ω, paired comparisons, (10) no simultaneous charge and discharge, (11) summary.

8 Code added for stage 1

- `cases.stage1_case(smax_each, w)`: the system above.
- `ErrorModel.sigma_w` and `forecasts.terminal_values`: the terminal-value error.
- `RunSettings.seed_w(d)`: its own seed stream.
- `runner.run_draw`: draws all owners' price forecasts together (so χ is shared), draws $\hat{w}$, builds `B_fcst` and `B_priv` with it; rows gain `sigma_w` and `w_hat_e`.
- `scripts/make_stage1_figures.py`: every figure and number in this document.

Stage 0 is unaffected: rerunning all 1 506 stage-0 draws reproduces `results/stage0/` to 10^{-12}, and all 21 tests pass (two new: the two-battery gate and the $\hat{w}$ error).